

Chapter 3 Discrete Random Variables and Probability Distributions

Introduction

We have a rule that assigns s to each simple event in S a single real number. Mathematically speaking, we are dealing with a function.

3.1 Random Variable

Random variable(RV)

For a given sample space S of some experiment, a random variable is any rule that associates a number with each outcome in S .

Or, a random variable is a function whose domain is the sample space and whose range is the set of real number.(RV is a function rather than variable)

Random variables are denoted by uppercase letters(X, Y , etc.). We use lowercase letters(x, y , etc.) to denote a variable. Here we use lowercase letters to represent some particular value of the corresponding random variable.

The notation $X(s) = x$ means that x is the value associated with the outcome s by the RV X .

Two Types of Random Variables

- Discrete Random Variable

Possible values either constitute a finite set or can be listed in an infinite sequence.

- Continuous Random Variable

Its set of possible values consists on an entire interval on the number line. And no way to create an infinite listing them.

Bernoulli Random Variable

Any random variable whose only possible are 0 and 1 is called Bernoulli random variable.

3.2 Probability Distributions for Discrete Random Variables

Probability Distribution

The probability distribution or probability mass function(pmf) of a discrete RV is defined for every number x by

$$p(x) = P(X = x) = P(\text{all } s \text{ in } S : X(s) = x)$$

Note:

$$p_i = p(x_i) \geq 0, \sum_{i=1}^n p(x_i) = 1$$

x	x_1	$\cdots$	x_n
$p(x)$	p_1	$\cdots$	p_n

A parameter of a Probability Distribution

Suppose $p(x)$ depends on a quantity that can be assigned any one of a number of possible values with each different value determining a different probability distribution. Such a quantity is called **parameter** of the distribution.

For example, $p(x_1) = \alpha, p(x_2) = 1 - \alpha$ where α is a parameter.

The collection of all probability distributions for different values of the parameter is called **a family of probability distributions**.

Cumulative Distribution Function

The cumulative distribution function(cdf) $F(x)$ of a discrete RV X with pmf $p(x)$ is defined for every number x by

$$F(x) = P(X \leq x) = \sum_{y: y \leq x} p(y)$$

- Proposition

For any two numbers a and b with $a \leq b$,

$$P(a \leq X \leq b) = F(b) - F(a-)$$

where $a-$ represents strictly less than a .

If the only possible values are integers and if a, b are integers, then

$$P(a \leq X \leq b) = F(b) - F(a - 1)$$

If $a = b$, we have $P(X = a) = F(a) - F(a - 1)$.

- Three Properties of cdf(discrete/continuous cases)

- Non-decreasing. i.e., $x_1 < x_2 \Leftrightarrow F(x_1) \leq F(x_2)$
- $F(-\infty) = \lim_{x \rightarrow -\infty} F(x) = 0, F(+\infty) = \lim_{x \rightarrow +\infty} F(x) = 1$
- $F(x + 0) = F(x)$

3.3 Expected Values of Discrete Random Variables

The expected value of X

Let X be a discrete RV with set of possible values D and pmf $p(x)$. The expected value or mean value of X , denoted by $E(X)$ or μ_X is

$$E(X) = \mu_X = \sum_{x \in D} x \cdot p(x)$$

If the sum does not exist, we say the expectation of X does not exist(infinite case)

For Bernoulli RV, $E(X) = p$.

- Example 3.19 find the expected value of $p(x) = p(1-p)^{x-1}$, $x = 1, 2, \dots$

$$\begin{aligned} E(x) &= \sum_{x=1}^{\infty} xp(1-p)^{x-1} \\ &= p \sum_{x=1}^{\infty} \left[-\frac{d}{dp} (1-p)^x \right] \\ &= -p \frac{d}{dp} \sum_{x=1}^{\infty} (1-p)^x \\ &= -p \cdot \frac{d}{dp} \left(\frac{1}{1-(1-p)} - 1 \right) \\ &= \frac{1}{p} \end{aligned}$$

The expected value of a function

Let X be a discrete RV with set of possible values D and pmf $p(x)$. The expected value or mean value of any function $h(X)$, denoted by $E[h(X)]$ or $\mu_{h(X)}$, is computed by

$$E[h(X)] = \sum_{x \in D} h(x) \cdot p(x)$$

- Rule of Expected Value

$$E(aX + b) = aE(X) + b$$

The Variance of X

Let X have pmf $p(x)$ and the expected value μ , then the variance of X denoted by $V(X)$ or σ_X^2 , or just σ^2 , is

$$V(X) = \sum_{x \in D} (x - \mu)^2 \cdot p(x) = E[(X - \mu)^2]$$

The standard deviation(SD) of X is

$$\sigma_X = \sqrt{\sigma_X^2}$$

- A short formula for σ^2

$$V(X) = \sigma^2 = \sum_D x^2 p(x) - \mu^2 = E(X^2) - [E(X)]^2$$

- Rules of Variance
 - $V(aX + b) = a^2 \sigma_X^2$
It is not related to b .

3.4 The Binomial Probability Distribution

Requirements for a binomial experiment

- The experiment consists of n smaller experiments called trials, where n is fixed in advance.
- Only one of two outcomes for each trail, which we denote by success(S) or failure(F).
- The trails are independent
- The probability of success is constant, denoted by p .
- Rule

If the sample size n is at most 5% of the population size, the experiment can be analyzed as though it were exactly a binomial experiment for experiment without replacement.

Binomial random variable

- Definition:
Given a binomial experiment consisting of n trails, the binomial random variable X associated with this experiment is defined as X is the number of S 's among the n trials.
- $X \sim \text{Bin}(n, p)$
Possible values for X in an n -trial experiment are $x = 0, 1, \dots, n$. We write $X \sim \text{Bin}(n, p)$ to indicate X is a binomial RV based on n trials with success probability p .
And because of the pmf of a binomial rv depends on the two parameters n and p . We denote the pmf by $b(x; n, p)$.
- Theorem:

$$b(x; n, p) = \begin{cases} \binom{n}{x} p^x (1-p)^{n-x}, & x = 0, 1, 2, \dots, n \\ 0, & \text{otherwise} \end{cases}$$

- Notation
For $X \sim \text{Bin}(n, p)$, the cdf will be denoted by

$$P(X \leq x) = B(x; n, p) = \sum_{y=0}^x b(y; n, p), x = 0, 1, \dots, n$$

- The mean and variance of X

If $X \sim \text{Bin}(n, p)$, then

$$E(X) = np, V(X) = np(1 - p) = npq$$

where $q = 1 - p$

The Geometric Distribution(Supplementary)

- Definition

The number of trials up to and including the **first success** in a sequence of independent Bernoulli trials with a constant success probability p has a **geometric distribution** with parameter p . The probability mass function is

$$P(X = x) = (1 - p)^{x-1}p$$

And

$$P(X \leq x) = 1 - (1 - p)^x$$

(It can be obtained by geometric progression)

And

$$E(X) = \frac{1}{p}, Var(X) = \frac{1 - p}{p^2}$$

3.5 Hypergeometric and Negative Binomial Distributions

- Proposition:

If **X 's is the number of S 's** in a completely random **sample of size n** drawn from a population consisting of M S 's and $(N - M)$ F 's, then the probability distribution of X called the **hypergeometric distribution** is given by

$$P(X = x) = h(x; n, M, N) = \frac{\binom{M}{x} \binom{N-M}{n-x}}{\binom{N}{n}}$$

- The range of rv X

$$\text{Max}(0, n - (N - M)) \leq X \leq \text{Min}(n, M)$$

- Proposition

The **mean and variance** of the hypergeometric rv X are

$$E(X) = np, V(X) = \frac{N - n}{N - 1} \cdot n \cdot p \cdot (1 - p)$$

where $p = \frac{M}{N}$ and $N - n/N - 1$ is less or equal to 1, called **finite population correction factor**.

The Negative Binomial Distribution

The negative binomial rv and distribution are based on an experiment satisfying the following conditions.

- The experiment consists of a sequence of independent trials
- Each trial can result in either a success or a failure
- The probability of success is constant
- The experiment continues **until a total of r successes have been observed**, where r is a **specified positive integer**.

The random variable is $X =$ **the number of failures** that precede the r th success. X is called **negative binomial variable**.

$$nb(x; r, p) = \binom{x + r - 1}{r - 1} p^r (1 - p)^x$$

For The first $x + r - 1$ trials, arrange $r - 1$ S . For the last, it is fixed S .

Thus

$$nb(x; r, p) = \binom{x + r - 1}{r - 1} p^{r-1} (1 - p)^x \cdot p = \binom{x + r - 1}{r - 1} p^r (1 - p)^x$$

In some sources, the negative binomial rv is taken to be the number of $X + r$ rather than the number of failures.

In the special case $r = 1$, the pmf is

$$nb(x; 1, p) = (1 - p)^x p, x = 0, 1, 2, \dots$$

Both $X =$ the number of F 's and $Y =$ number of trials ($1 + X$) are called **geometric random variables**. The pmf is called the **geometric distribution**.

Proposition:

If X is a negative binomial rv, then

$$E(X) = \frac{r(1 - p)}{p}, V(X) = \frac{r(1 - p)}{p^2}$$

3.6 The Poisson Probability Distribution

Poisson Distribution

A random variable X is said to have a **Poisson distribution** with parameter $\lambda (\lambda > 0)$ if the pmf of X is

$$p(x; \lambda) = \frac{e^{-\lambda} \lambda^x}{x!},$$

The value of λ is **frequency a rate per unit time or per unit area**. (mean)

The Maclaurin infinite series expansion of e^λ :

$$e^\lambda = 1 + \lambda + \frac{\lambda^2}{2!} + \dots = \sum_{x=0}^{\infty} \frac{\lambda^x}{x!}$$

Thus

$$\sum_{n=0}^{\infty} e^{-\lambda} \frac{\lambda^x}{x!} = 1$$

Proposition:

If X has a Poisson distribution with parameter λ , then

$$E(X) = V(x) = \lambda$$

The Poisson Distribution as a Limit

The rationale for using the Poisson distribution in many situations is provided by the following proposition

Proposition:

Suppose that in the **binomial pmf** $b(x; n, p)$, we let $n \rightarrow \infty$ and $p \rightarrow 0$ in such a way that **np approaches a value $\lambda > 0$** , then $b(x; n, p) \rightarrow p(x; \lambda)$.

That is, when n is large and p is small, $b(x; n, p) \approx p(x; \lambda)$. This approximation can safely be applied if $n \geq 100$, $p \leq 0.01$ and $np \leq 20$.